Simulating Clifford+T Circuits via Generalized Stabilizer States

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Abstract

This document establishes the formal framework for simulating Clifford+T circuits using generalized stabilizer states. By extending the reference frame and parity tracking approach of Stim, we introduce a theoretical partition between geometric basis transformations and probabilistic state evolution. We prove that the superposition dimension and basis alignment of the active state can be evaluated ahead of time, independent of probabilistic runtime collapse. This natural decoupling motivates an offline-compilation and fast-sampling architecture (e.g., the HIR compiler and SchrodingerVM of UCC), where heavy geometric basis alignments are evaluated offline, allowing the complex probability amplitudes to execute over minimal, dense state arrays using optimized division-free operations.

1 The Circuit Model

A quantum circuit over n qubits is a series of operations $O^{(t)}$ applied at discrete time steps $t = 0, \ldots$. At $t = 0$ qubits are in the computational basis state $|0\rangle^{\otimes n}$. Let $P \in \mathcal{P}_n$ denote physical n-qubit Pauli string operators, with $\mathcal{P}_n = \{\pm 1, \pm i\}\{I, X, Y, Z\}^{\otimes n}$ representing the n-qubit Pauli group.

Each operation is one of the following types:

- **Clifford Unitaries (U):** $U \in \mathcal{C}_n$. The Clifford group is $\mathcal{C}_n = \{U \in U(2^n) : U\mathcal{P}_n U^\dagger = \mathcal{P}_n\}$—the set of gates that maps Paulis to Paulis. $\{H, S, CNOT\}$ is a common generating set for $\mathcal{C}_n$.
- **Non-Clifford Unitaries (e.g., T_q):** $T_q = e^{i\pi/8}\cos\left(\frac{\pi}{8}\right)[I - i\tan\left(\frac{\pi}{8}\right)Z_q]$. We will focus on T although the formalism generalizes to any unitary generated by a Pauli operator.
- **Projective Pauli Measurements (P):** We measure $P \in \mathcal{P}_n$. It has outcomes $m \in \{0, 1\}$ corresponding to the observed eigenvalue $(-1)^m$, with projectors $\Pi_m = \frac{1}{2}(I + (-1)^m P)$.
- **Stochastic Pauli Noise (E):** $E \in \mathcal{P}_n$ is a Pauli error that occurs with some probability p_E.
- **Classically Controlled Paulis (e.g., $P^{m^{(k)}}$):** Applies $P \in \mathcal{P}_n$ conditioned on a classical measurement outcome $m^{(k)}$ recorded at an earlier step k.

It will be useful to assume that all operations are single-qubit or two-qubit gates, as is common in quantum circuits.

Although there are valid operations outside this set, this model is universal for quantum computing. The remainder of this document focuses on the theoretical framework for simulating circuits composed of these operations.

2 Stabilizer-Destabilizer Tableaus

- $\mathcal{S} \subset \mathcal{P}_n$ is a stabilizer group which stabilizes the n-qubit stabilizer state $|\psi_{\mathcal{S}}\rangle$ with $+1$ eigenvalue. The group does not contain $-I$ to avoid being trivial.
- $\mathcal{S} = \langle s_1, \ldots, s_n \rangle$ is generated by n independent commuting signed Pauli strings s_i, where $s_i |\psi_{\mathcal{S}}\rangle = |\psi_{\mathcal{S}}\rangle$ for all i.
- $\mathcal{D} = \langle d_1, \ldots, d_n \rangle$ is the destabilizer group generated by n independent signed Pauli strings d_i that anticommute with their corresponding stabilizer generators: $\{d_i, s_i\} = 0$ and commute with all others: $[d_i, s_j] = 0$ for $j \neq i$, that is $d_i s_j = (-1)^{\delta_{ij}} s_j d_i$.
- In the canonical form, destabilizers mutually commute: $[d_i, d_j] = 0$.
- Define the (signed) Pauli $\mathcal{T} = (\mathcal{S}, \mathcal{D})$ as the joint stabilizer/destabilizer pair along with their signs.
- $\mathcal{T}$ serves as a basis for the Pauli group, where we define the extraction as a grouped product:

$$P = \omega \left(\prod_{i=1}^{n} d_i^{x_i} \right) \left(\prod_{i=1}^{n} s_i^{z_i} \right) = \omega \mathbf{d}^{\mathbf{x}} \mathbf{s}^{\mathbf{z}} \tag{1}$$

  where the product is grouped such that all destabilizers precede all stabilizers to consistently fix the phase ω, with $\omega \in \{\pm 1, \pm i\}$ and the bitwise vectors $\mathbf{x}, \mathbf{z} \in \mathbb{F}_2^n$[1] indicating the presence of the corresponding generators in the decomposition of P.
- Let $\Gamma(P, \mathcal{T}) \mapsto (\omega, \mathbf{x}, \mathbf{z})$ be the *Pauli Extraction Map* that extracts this decomposition.
- Let $\Lambda_C(U, \mathcal{T}) \mapsto \mathcal{T}'$ represent the *Clifford Conjugation* operation that updates the tableau by conjugating all generators by a physical Clifford unitary U, i.e. $\mathcal{T}' = (U\mathcal{S}U^\dagger, U\mathcal{D}U^\dagger)$. This is a symplectic transformation, meaning it preserves all canonical commutation relations and rotates the geometry of the tableau.
- Let $\Lambda_{\text{rowsum}}(i, j, \mathcal{T}) \mapsto \mathcal{T}'$ represent a *Symplectic Rowsum*. This operation performs a change-of-basis by multiplying source stabilizer j into target stabilizer i, replacing $s_i \leftarrow s_i s_j$. To preserve canonical commutation relations (preventing d_j from anticommuting with the new s_i), the destabilizer must concurrently update via $d_j \leftarrow d_j d_i$. Because s_i and s_j both stabilize the vacuum state, their product $s_i s_j$ also stabilizes it. This acts as a structural gauge choice to reshape the tableau's geometry without altering the physical vacuum state.
- Let $\Lambda_{\text{pivot}}(k_p, \omega', \mathbf{z}', \mathcal{T}) \mapsto \mathcal{T}'$ represent the *Aaronson-Gottesman (AG) Pivot*[2] operation that performs a change-of-basis by swapping generators and restoring canonical commutation relations for a new observable P.

[1] Throughout the rest of the document, we will use boldface lowercase letters to represent these bitwise vectors for compactness. We define the standard basis vector $\mathbf{e}_i \in \mathbb{F}_2^n$ (where the i-th bit is 1 and all others 0), the bitwise dot product $\mathbf{b}_1 \cdot \mathbf{b}_2 = \sum_{i=1}^{n} b_{1,i} b_{2,i} \pmod 2$, the bitwise XOR $(\mathbf{b}_1 \oplus \mathbf{b}_2)_i = b_{1,i} \oplus b_{2,i} \pmod 2$, the bitwise AND $(\mathbf{b}_1 \wedge \mathbf{b}_2)_i = b_{1,i} \cdot b_{2,i} \pmod 2$, the bitwise OR $(\mathbf{b}_1 \vee \mathbf{b}_2)_i = b_{1,i} \vee b_{2,i}$, and the bitwise NOT $(\neg \mathbf{b})_i = 1 - b_i \pmod 2$. The subset relation $\boldsymbol{\alpha} \subseteq \mathbf{b}$ denotes bitwise inclusion ($\boldsymbol{\alpha} \wedge \neg \mathbf{b} = \mathbf{0}$). We use $\boldsymbol{\alpha}$ to represent indexing sub-masks associated with branches of a superposition.

[2] In quantum literature, the overall measurement resolution process is frequently referred to as symplectic Gaussian elimination. Strictly speaking, Gaussian elimination specifically denotes the prerequisite $\mathcal{O}(n^2)$ preparation phase of iterative rowsums, while the AG Pivot is the subsequent $\mathcal{O}(1)$ action of swapping the generators ($s_k \leftrightarrow d_k$) and phase anchoring.

This happens post measurement, when the state collapses for a physical measurement $P \in \mathcal{P}_n$. If P commutes with all stabilizers, this is a no-operation. Assuming anticommutativity has already been cleared from all stabilizers except a chosen pivot s_{k_p} (via prerequisite rowsums, which guarantees the extracted X-support $\mathbf{x}$ is exactly $\mathbf{e}_{k_p}$), the AG Pivot operation requires the pivot axis k_p, the phase ω', and the current Z-support $\mathbf{z}'$. It demotes s_{k_p} to the new destabilizer d_{k_p} and sets the new s_{k_p} to P, handling phase housekeeping to maintain canonical form.

- Following the insight of Stim, we maintain both a forward tableau $\mathcal{T}_{fwd}$ and an inverse tableau $\mathcal{T}_{inv}$. The forward tableau defines the current generators in terms of physical Paulis, while the inverse tableau defines physical Paulis in the current stabilizer basis. For the local, constant-weight observables typical of quantum circuits, the inverse tableau can evaluate the Pauli extraction map using $\mathcal{O}(n)$ bitwise operations (executed as highly-optimized, contiguous SIMD vector reads) by simply reading and XORing the corresponding rows of $\mathcal{T}_{inv}$. In contrast, extracting this using only the forward tableau would require an $\mathcal{O}(n^2)$ generic inner product against all $2n$ rows, or—if optimized for local observables—an $\mathcal{O}(n)$ sequence of scattered, cache-unfriendly scalar column-extractions that defeats vectorization. Given that the Pauli extraction map is used in every operation that interacts with the physical hardware (measurements, non-Clifford gates, noise), this is a critical optimization.
- Clifford conjugation scales as $\mathcal{O}(n)$ vector operations for low-arity gates.
- Symplectic Rowsums scale as $\mathcal{O}(n)$ vector operations since one stabilizer and one destabilizer are updated in both tableaus.
- The overall AG Pivot process (including the prerequisite Gaussian elimination) scales as $\mathcal{O}(n^2)$ vector operations because it requires manipulating the entire tableau.

Need to revisit all these complexities in the main text/for the operations instead

3 The Unified State Tuple and Symplectic Determinism

In addition to being a basis for the Pauli group, the tableau $\mathcal{T}$ induces a basis for the entire Hilbert space of n-qubits. The destabilizers act as ladder operators that shift the active basis state by toggling the presence of their corresponding operator in the product, while the stabilizers define the vacuum state $|\psi_{\mathcal{S}}\rangle$. To make this perspective, clear, we will switch notation to $|\mathbf{0}_{\mathcal{S}}\rangle$ for the state stabilized by $\mathcal{S}$.

We partition the generalized state at time t into a discrete, unified tuple $\mathcal{X}^{(t)}$ composed of two domains: the geometric frame and the probabilistic state.

$$\mathcal{X}^{(t)} := \Big(\underbrace{\mathcal{T}^{(t)}, \mathbf{b}^{(t)}}_{\text{Geometric State}} \quad , \quad \underbrace{\mathcal{M}^{(t)}, \gamma^{(t)}, \mathbf{p}_X^{(t)}, \mathbf{p}_Z^{(t)}, a^{(t)}}_{\text{Probabilistic State}} \Big) \tag{2}$$

where:

- $\mathcal{T}^{(t)} \equiv (\mathcal{S}^{(t)}, \mathcal{D}^{(t)})$ is the fully signed geometric frame.
- $\mathbf{b}^{(t)} \in \mathbb{F}_2^n$ is the *active support mask*. It isolates which destabilizers are currently active for at least some non-zero terms in the superposition. For all indices where $b_i^{(t)} = 0$, the corresponding generator d_i is dormant.

- $\mathcal{M}^{(t)}$ is the classical memory record, defined as an ordered sequence tracking all historical binary measurement outcomes. We use $\langle\rangle$ to denote an empty sequence and $\|$ for sequence concatenation.

- $\gamma^{(t)} \in \mathbb{C}$ is a global complex scale factor tracking the physical norm and global continuous phase.

- $\mathbf{p}_X^{(t)}, \mathbf{p}_Z^{(t)} \in \mathbb{F}_2^n$ are purified classical parity trackers governing the dynamic *probabilistic deviations* (runtime physical signs).

- $a^{(t)} \in \mathbb{C}^{2^{|\mathbf{b}^{(t)}|}}$ is the complex amplitude array. It is indexed by the active branch sub-mask $\boldsymbol{\alpha} \subseteq \mathbf{b}^{(t)}$. Because $\boldsymbol{\alpha}_k = 0$ for all dormant bits $k \notin \mathbf{b}^{(t)}$, the dimension of this array is exactly $2^{|\mathbf{b}^{(t)}|}$. These entries are unnormalized; we rely on $\gamma^{(t)}$ to track the global norm and global phase, while $a^{(t)}$ tracks the relative complex structure of the superposition.

We assume $\xi^{(t)} \in [0, 1)$ is a globally available continuous random variable that we can query at any time step t to resolve stochastic outcomes.

For our initial state, we have $\mathcal{T}^{(0)}$ defined by the n-qubit computational basis state, $\mathbf{b}^{(0)} = \mathbf{0}$, $\mathcal{M}^{(0)} = \langle\rangle$, $\gamma^{(0)} = 1$, $\mathbf{p}_X^{(0)} = \mathbf{0}$, $\mathbf{p}_Z^{(0)} = \mathbf{0}$, and $a^{(0)} = [1]$.

The goal of this section is to formalize how these state variables change for each circuit operation. We will find that the geometric state rules are independent of the probabilistic state, enabling the decoupling of offline geometric compilation from online probability sampling.

3.1 The Physical State Equation

To recover the physical quantum state $|\psi_{\mathcal{X}^{(t)}}\rangle$ from the unified tuple $\mathcal{X}^{(t)}$, we define the *true physical generators* by attaching the runtime probabilistic parities to the geometric generators:

$$\bar{d}_i^{(t)} = (-1)^{p_{X,i}^{(t)}} d_i^{(t)} \quad \text{and} \quad \bar{s}_i^{(t)} = (-1)^{p_{Z,i}^{(t)}} s_i^{(t)} \tag{3}$$

The physical base state/vacuum state $\left|\mathbf{0}_{\mathcal{S}^{(t)}}(\mathbf{p}_Z^{(t)})\right\rangle$ is defined as the simultaneous $+1$ eigenstate of the physical stabilizers. The full physical state is then constructed by applying the complex amplitudes over the active sub-basis defined by the mask $\mathbf{b}^{(t)}$ (summing over bitstrings $\boldsymbol{\alpha}$ where $\boldsymbol{\alpha}_i = 0$ for all $b_i^{(t)} = 0$):

$$|\psi_{\mathcal{X}^{(t)}}\rangle = \gamma^{(t)} \sum_{\boldsymbol{\alpha} \subseteq \mathbf{b}^{(t)}} a_{\boldsymbol{\alpha}}^{(t)} (\bar{\mathbf{d}}^{(t)})^{\boldsymbol{\alpha}} \left|\mathbf{0}_{\mathcal{S}^{(t)}}(\mathbf{p}_Z^{(t)})\right\rangle \tag{4}$$

where we use the vector exponentiation notation $(\bar{\mathbf{d}}^{(t)})^{\boldsymbol{\alpha}} \equiv \prod_{i=1}^n (\bar{d}_i^{(t)})^{\boldsymbol{\alpha}_i}$. In some derivations, we will expand the physical generators which yields $(\bar{\mathbf{d}}^{(t)})^{\boldsymbol{\alpha}} = (-1)^{\mathbf{p}_X^{(t)} \cdot \boldsymbol{\alpha}} (\mathbf{d}^{(t)})^{\boldsymbol{\alpha}}$.

3.2 Tableau Operations on the State

3.2.1 Pauli Action

For all operations aside from Clifford gates, we will need to consider the action of a localized pauli $P^{(t)}$ action on the current state. From the Pauli extraction map $\Gamma(P^{(t)}, \mathcal{T}^{(t)}) \mapsto (\omega, \mathbf{x}, \mathbf{z})$, which means we can write

$$P^{(t)} = \omega(\mathbf{d}^{(t)})^{\mathbf{x}} (\mathbf{s}^{(t)})^{\mathbf{z}} = \omega(-1)^{\mathbf{x} \cdot \mathbf{p}_X^{(t)} + \mathbf{z} \cdot \mathbf{p}_Z^{(t)}} (\bar{\mathbf{d}}^{(t)})^{\mathbf{x}} (\bar{\mathbf{s}}^{(t)})^{\mathbf{z}} \tag{5}$$

where we've expanded using the physical stabilizers and destabilizers.

Applying this to Eq. 4 yields:

$$
\begin{aligned}
P^{(t)}\left|\psi_{\mathcal{X}^{(t)}}\right\rangle &= \gamma^{(t)} \sum_{\boldsymbol{\alpha}\subseteq \mathbf{b}^{(t)}} a_{\boldsymbol{\alpha}}^{(t)} \Big(\omega(-1)^{\mathbf{x}\cdot\mathbf{p}_X^{(t)}+\mathbf{z}\cdot\mathbf{p}_Z^{(t)}} (\bar{\mathbf{d}}^{(t)})^{\mathbf{x}} (\bar{\mathbf{s}}^{(t)})^{\mathbf{z}}\Big) (\bar{\mathbf{d}}^{(t)})^{\boldsymbol{\alpha}} \left|\mathbf{0}_{\mathcal{S}^{(t)}}(\mathbf{p}_Z^{(t)})\right\rangle \\
&= \gamma^{(t)} \sum_{\boldsymbol{\alpha}\subseteq \mathbf{b}^{(t)}} a_{\boldsymbol{\alpha}}^{(t)} \omega(-1)^{\mathbf{x}\cdot\mathbf{p}_X^{(t)}+\mathbf{z}\cdot\mathbf{p}_Z^{(t)}} (-1)^{\mathbf{z}\cdot\boldsymbol{\alpha}} (\bar{\mathbf{d}}^{(t)})^{\mathbf{x}} (\bar{\mathbf{d}}^{(t)})^{\boldsymbol{\alpha}} (\bar{\mathbf{s}}^{(t)})^{\mathbf{z}} \left|\mathbf{0}_{\mathcal{S}^{(t)}}(\mathbf{p}_Z^{(t)})\right\rangle \quad \text{(Commute } \bar{\mathbf{s}} \text{ past } \bar{\mathbf{d}}\text{)} \\
&= \gamma^{(t)} \sum_{\boldsymbol{\alpha}\subseteq \mathbf{b}^{(t)}} a_{\boldsymbol{\alpha}}^{(t)} \underbrace{\omega(-1)^{\mathbf{x}\cdot\mathbf{p}_X^{(t)}+\mathbf{z}\cdot\mathbf{p}_Z^{(t)}+\mathbf{z}\cdot\boldsymbol{\alpha}}}_{\text{Dynamic Phase: } \Phi(\boldsymbol{\alpha};\omega,\mathbf{x},\mathbf{z},\mathbf{p}_X^{(t)},\mathbf{p}_Z^{(t)})} \left((\bar{\mathbf{d}}^{(t)})^{\boldsymbol{\alpha}\oplus\mathbf{x}} \left|\mathbf{0}_{\mathcal{S}^{(t)}}(\mathbf{p}_Z^{(t)})\right\rangle\right) \quad \text{(Evaluate } \bar{\mathbf{s}} \text{ on vacuum)}
\end{aligned}
\tag{6}
$$

where we have defined the dynamic phase as

$$
\Phi(\boldsymbol{\alpha};\omega,\mathbf{x},\mathbf{z},\mathbf{p}_X^{(t)},\mathbf{p}_Z^{(t)}) = \omega(-1)^{\mathbf{x}\cdot\mathbf{p}_X^{(t)}+\mathbf{z}\cdot\mathbf{p}_Z^{(t)}+\mathbf{z}\cdot\boldsymbol{\alpha}} \tag{7}
$$

3.2.2 Rowsums and State Evolution

When applying a Symplectic Rowsum $\Lambda_{\text{rowsum}}(i,j)$, we reshape the geometry by multiplying source stabilizer j into target stabilizer i. To preserve canonical commutativity, the corresponding destabilizers must update inversely.

Let the tilde ($\sim$) denote variables within the *new* mapped geometric basis. The rowsum enforces the following transformation:

$$
\begin{aligned}
\tilde{s}_i = s_i s_j \quad &\Longrightarrow \quad s_i = \tilde{s}_i \tilde{s}_j \\
\tilde{d}_j = d_j d_i \quad &\Longrightarrow \quad d_j = \tilde{d}_j \tilde{d}_i
\end{aligned}
$$

All other generators remain unchanged.

Evolution of the Physical State

The physical quantum state $|\psi_{\mathcal{X}^{(t)}}\rangle$ (from Eq. 4) must remain invariant beneath this mutating geometry. We derive the new state representation by substituting the inverted basis relationships into the state equation.

First, we determine the new runtime Z-parities that define the new physical vacuum state $\left|\mathbf{0}_{\tilde{\mathcal{S}}}\right\rangle$. Because $\tilde{s}_i = s_i s_j$, the new physical stabilizer inherits the product of the old eigenvalues:

$$
(-1)^{\tilde{p}_{Z,i}} = (-1)^{p_{Z,i}} (-1)^{p_{Z,j}} = (-1)^{p_{Z,i}\oplus p_{Z,j}} \tag{8}
$$

Next, we substitute the inverted destabilizer relationship $d_j = \tilde{d}_j \tilde{d}_i$ into the geometric product $(\mathbf{d})^{\boldsymbol{\alpha}}$ for a given active branch $\boldsymbol{\alpha}$:

$$
(d_i)^{\boldsymbol{\alpha}_i} (d_j)^{\boldsymbol{\alpha}_j} = (\tilde{d}_i)^{\boldsymbol{\alpha}_i} (\tilde{d}_j \tilde{d}_i)^{\boldsymbol{\alpha}_j} = (\tilde{d}_i)^{\boldsymbol{\alpha}_i \oplus \boldsymbol{\alpha}_j} (\tilde{d}_j)^{\boldsymbol{\alpha}_j}
$$

This reveals that the old basis configuration $\boldsymbol{\alpha}$ maps to a new configuration $\tilde{\boldsymbol{\alpha}}$ where the i-th bit has been flipped by the j-th bit: $\tilde{\boldsymbol{\alpha}}_i = \boldsymbol{\alpha}_i \oplus \boldsymbol{\alpha}_j$, which we write as $\tilde{\boldsymbol{\alpha}} = \boldsymbol{\alpha} \oplus (\boldsymbol{\alpha}_j \cdot \mathbf{e}_i)$. Because this coordinate shift evaluates as an involution over $\mathbb{F}_2$ (applying it twice yields the original index), the inverse mapping is identical: $\boldsymbol{\alpha} = \tilde{\boldsymbol{\alpha}} \oplus (\tilde{\boldsymbol{\alpha}}_j \cdot \mathbf{e}_i)$.

Finally, we substitute this index shift into the runtime X-parity phase term $(-1)^{\mathbf{p}_X \cdot \boldsymbol{\alpha}}$ to isolate the new tracking vector $\tilde{\mathbf{p}}_X$:

$$
\mathbf{p}_X \cdot \boldsymbol{\alpha} = \mathbf{p}_X \cdot \big(\tilde{\boldsymbol{\alpha}} \oplus (\tilde{\boldsymbol{\alpha}}_j \cdot \mathbf{e}_i)\big) = (\mathbf{p}_X \cdot \tilde{\boldsymbol{\alpha}}) \oplus (p_{X,i} \cdot \tilde{\boldsymbol{\alpha}}_j)
$$

Grouping the terms acting on $\tilde{\boldsymbol{\alpha}}_j$ yields the new X-parity: $\tilde{p}_{X,j} = p_{X,j} \oplus p_{X,i}$.

Explicit Update Rules ($\mathcal{X} \mapsto \tilde{\mathcal{X}}$):

To ensure the state remains invariant, the complex amplitudes and physical parity trackers must absorb these coordinate mappings. A geometric rowsum forces the following state tuple update:

$$\begin{aligned}
\tilde{\mathcal{T}} &= \Lambda_{\text{rowsum}}(i, j, \mathcal{T}) && \text{(Geometric Update)} \\
\tilde{\mathbf{b}} &= \mathbf{b}, \quad \tilde{\gamma} = \gamma && \text{(Invariants)} \\
\tilde{p}_{Z,i} &= p_{Z,i} \oplus p_{Z,j} && \text{(Vacuum Alignment)} \\
\tilde{p}_{X,j} &= p_{X,j} \oplus p_{X,i} && \text{(Phase Alignment)} \\
\tilde{a}_{\boldsymbol{\beta}} &= a_{\boldsymbol{\beta} \oplus (\boldsymbol{\beta}_j \cdot \mathbf{e}_i)} \quad \forall \boldsymbol{\beta} \subseteq \tilde{\mathbf{b}} && \text{(Amplitude Reindexing Map)}
\end{aligned} \tag{9}$$

(Note: we introduce the dummy index $\boldsymbol{\beta}$ to represent indices within the newly mapped array $\tilde{a}$).

The Zero-Cost Dormant Property:

A critical algorithmic optimization emerges from the reindexing map if the source axis of the rowsum is dormant ($j \notin \mathbf{b}^{(t)}$). Because the active support mask dictates that $\boldsymbol{\beta}_j = 0$ across all valid branches of the superposition, the coordinate shift evaluates to zero: $\boldsymbol{\beta} \oplus \mathbf{0} = \boldsymbol{\beta}$. When evaluating a rowsum from a dormant source, the complex array avoids the Amplitude Reindexing Map and passes through untouched. Note that the parity shifts $(\tilde{\mathbf{p}}_Z, \tilde{\mathbf{p}}_X)$ must still be dynamically evaluated to preserve the canonical vacuum phase.

4 Unified State Evolution ($\mathcal{X}^{(t)} \mapsto \mathcal{X}^{(t+1)}$)

Every circuit operation $O^{(t)}$ maps $\mathcal{X}^{(t)} \mapsto \mathcal{X}^{(t+1)}$. In this section, we derive the explicit update rules for each operation type. If a state member is not listed in a transition rule, it remains invariant.

4.1 Cliffords

Derivation: From Clifford Conjugation, U maps the tableau $\mathcal{T}^{(t)}$ to $\mathcal{T}' = \Lambda_C(U, \mathcal{T}^{(t)})$ by conjugating all generators. The vacuum eigenstate condition satisfies the new base state $\left|\mathbf{0}_{\mathcal{S}^{(t)}}(\mathbf{p}_Z^{(t)})\right\rangle$:

$$\begin{aligned}
\bar{s}_i^{(t+1)} \left(U \left|\mathbf{0}_{\mathcal{S}^{(t)}}(\mathbf{p}_Z^{(t)})\right\rangle\right) &= U \bar{s}_i^{(t)} U^\dagger U \left|\mathbf{0}_{\mathcal{S}^{(t)}}(\mathbf{p}_Z^{(t)})\right\rangle \\
&= U\big(\bar{s}_i^{(t)} \left|\mathbf{0}_{\mathcal{S}^{(t)}}(\mathbf{p}_Z^{(t)})\right\rangle\big) \\
&= (+1) \left(U \left|\mathbf{0}_{\mathcal{S}^{(t)}}(\mathbf{p}_Z^{(t)})\right\rangle\right)
\end{aligned}$$

Because symplectic tableau conjugation drops the global phase of U, we explicitly accumulate the global phase scalar θ_U acquired during the operation. Applying U to Eq. 4 yields:

$$U \left|\psi_{\mathcal{X}^{(t)}}\right\rangle = (\gamma^{(t)} \cdot e^{i\theta_U}) \sum_{\boldsymbol{\alpha} \subseteq \mathbf{b}^{(t)}} a_{\boldsymbol{\alpha}}^{(t)} \, (\bar{\mathbf{d}}^{(t+1)})^{\boldsymbol{\alpha}} \left|\mathbf{0}_{\mathcal{S}^{(t)}}(\mathbf{p}_Z^{(t)})\right\rangle$$

Explicit Update Rules ($\mathcal{X}^{(t)} \mapsto \mathcal{X}^{(t+1)}$):

$$\begin{aligned}
\mathcal{T}^{(t+1)} &= \Lambda_C(U^{(t)}, \mathcal{T}^{(t)}) && \text{(Clifford Conjugation)} \\
\gamma^{(t+1)} &= \gamma^{(t)} \cdot e^{i\theta_U} && \text{(Global Phase Accumulation)}
\end{aligned}$$

The probabilistic state components $(\mathcal{M}, \mathbf{p}_X, \mathbf{p}_Z, a)$ remain invariant.

4.2 Non-Cliffords (e.g., T)

We evaluate non-Clifford unitaries expanded as sums over Pauli terms, focusing primarily on the T_q rotation: $T_q = e^{i\pi/8}\cos\left(\frac{\pi}{8}\right)[I - i\tan\left(\frac{\pi}{8}\right)Z_q]$ which operations on qubit q. The overall phase will get absorbed into γ and the I acts trivially, so our main task is to determine how Z_q acts.

Derivation: From the Pauli extraction map $\Gamma(Z_q^{(t)}, \mathcal{T}^{(t)}) \mapsto (\omega, \mathbf{x}, \mathbf{z})$, we geometrically expand the operator as $Z_q^{(t)} = \omega(\mathbf{d}^{(t)})^{\mathbf{x}}(\mathbf{s}^{(t)})^{\mathbf{z}}$. Let $\mathbf{x}_{\text{dorm}} = \mathbf{x} \wedge \neg\mathbf{b}^{(t)}$ be the mask of dormant *destabilizers* this operator has support on.

If there is more than one such dormant destabilizer, we isolate this support to a single geometric axis. We select a pivot $k_p \in \text{supp}(\mathbf{x}_{\text{dorm}})$ and iteratively apply the Symplectic Rowsum $\Lambda_{\text{rowsum}}(i, k_p)$ for all other $i \in \text{supp}(\mathbf{x}_{\text{dorm}}) \setminus \{k_p\}$. To understand why this clears the other dormant supports from Z_q, consider a case where Pauli extraction maps the product $d_i d_{k_p}$. Applying $\Lambda_{\text{rowsum}}(i, k_p)$ updates the geometry to $\tilde{d}_{k_p} = d_{k_p} d_i$, which inverts to $d_{k_p} = \tilde{d}_{k_p}\tilde{d}_i$ and $d_i = \tilde{d}_i$. Substituting these inverse maps into the operator yields:

$$d_i d_{k_p} \mapsto \tilde{d}_i(\tilde{d}_{k_p}\tilde{d}_i) = \tilde{d}_{k_p}(\tilde{d}_i)^2 = \tilde{d}_{k_p}$$

Because canonical destabilizers mutually commute and square to the identity over $\mathbb{F}_2$, the i-th destabilizer is eliminated from the geometric expansion. From Section 3.2.2, this intermediate geometric compression preserves the complex array invariant without reindexing a (the Zero-Cost Dormant Property).

Let $(\mathcal{T}_{\text{tmp}}, \mathbf{p}_X^{\text{tmp}}, \mathbf{p}_Z^{\text{tmp}})$ represent the intermediate geometric state and parities after these rowsums. We re-extract the expansion of Z_q in this new intermediate basis: $(\omega', \mathbf{x}', \mathbf{z}') = \Gamma(Z_q^{(t)}, \mathcal{T}_{\text{tmp}})$, restricting support to at most one dormant destabilizer. Following the Pauli action from Section 3.2.1, we see the action of Z_q in this intermediate basis is

$$\Phi(\boldsymbol{\alpha}; \omega', \mathbf{x}', \mathbf{z}', \mathbf{p}_X^{\text{tmp}}, \mathbf{p}_Z^{\text{tmp}})(\bar{\mathbf{d}}^{\text{tmp}})^{\boldsymbol{\alpha}\oplus\mathbf{x}'} \left|\mathbf{0}_{\mathcal{S}^{tmp}}(\mathbf{p}_Z^{tmp})\right\rangle$$

This logic maps the specific branch $\boldsymbol{\alpha}$ to interfere with the branch $\boldsymbol{\alpha} \oplus \mathbf{x}'$.

Explicit Update Rules ($\mathcal{X}^{(t)} \mapsto \mathcal{X}^{(t+1)}$):
We formalize the execution in three distinct phases:

Phase 1: Geometric Compression
If $|\mathbf{x}_{\text{dorm}}| > 1$, we select a source pivot $k_p \in \text{supp}(\mathbf{x}_{\text{dorm}})$ and perform $\Lambda_{\text{rowsum}}(i, k_p)$ for all $i \in \text{supp}(\mathbf{x}_{\text{dorm}}) \setminus \{k_p\}$. This yields the intermediate state $(\mathcal{T}_{\text{tmp}}, \mathbf{p}_X^{\text{tmp}}, \mathbf{p}_Z^{\text{tmp}})$. The Zero-Cost Dormant Property guarantees the array $a^{(t)}$ remains invariant. We re-extract the compressed operator $(\omega', \mathbf{x}', \mathbf{z}') = \Gamma(Z_q^{(t)}, \mathcal{T}_{\text{tmp}})$.

Phase 2: Dimensionality Expansion
Let $\mathbf{x}'_{\text{dorm}} = \mathbf{x}' \wedge \neg\mathbf{b}^{(t)}$ be the extracted dormant support. If the operator activates a dormant destabilizer ($\mathbf{x}'_{\text{dorm}} \neq \mathbf{0}$), the superposition dimension doubles: $\mathbf{b}^{(t+1)} = \mathbf{b}^{(t)} \vee \mathbf{x}'_{\text{dorm}}$. Let a^{pad} represent the array extended to the new domain $\mathbf{b}^{(t+1)}$, where all newly allocated branches ($\boldsymbol{\alpha} \subseteq \mathbf{b}^{(t+1)}$ where $\boldsymbol{\alpha} \not\subseteq \mathbf{b}^{(t)}$) are zero-padded. If no dormant destabilizer is activated, $\mathbf{b}^{(t+1)} = \mathbf{b}^{(t)}$ and $a^{\text{pad}} = a^{(t)}$.

Phase 3: Amplitude Mixing
Defining for compactness $\Phi'(\boldsymbol{\alpha}) := \Phi(\boldsymbol{\alpha}; \omega', \mathbf{x}', \mathbf{z}', \mathbf{p}_X^{\text{tmp}}, \mathbf{p}_Z^{\text{tmp}})$, we update the state:

$$\begin{aligned}
&\mathcal{T}^{(t+1)} = \mathcal{T}_{\text{tmp}}, \quad \mathbf{p}_X^{(t+1)} = \mathbf{p}_X^{\text{tmp}}, \quad \mathbf{p}_Z^{(t+1)} = \mathbf{p}_Z^{\text{tmp}} \\
&\gamma^{(t+1)} = \gamma^{(t)} \cdot e^{i\pi/8}\cos\left(\tfrac{\pi}{8}\right) \\
&a_{\boldsymbol{\alpha}}^{(t+1)} = a_{\boldsymbol{\alpha}}^{\text{pad}} - i\tan\left(\tfrac{\pi}{8}\right) \cdot \Phi'(\boldsymbol{\alpha} \oplus \mathbf{x}') \cdot a_{\boldsymbol{\alpha}\oplus\mathbf{x}'}^{\text{pad}} \quad \forall \boldsymbol{\alpha} \subseteq \mathbf{b}^{(t+1)}
\end{aligned}$$

4.3 Projective Pauli Measurements ($P^{(t)}$)

Deferred Normalization Framework:
To limit the need for runtime division, the array $a^{(t)}$ is unnormalized ($p_0 = \sum |a|^2 \neq 1$) while the overall factor $\gamma^{(t)}$ absorbs the overall normalizaiton changes. Thus to recover probability P_0, we have to multiple the unnormalized sum by this factor:

$$P_0 = p_0 \cdot |\gamma^{(t)}|^2 \tag{10}$$

Following a measurement collapse, we preserve this overall state norm ($|\gamma|^2 \sum |a|^2 = 1$) performing division exclusively on γ.

Derivation:

The absolute probability of observing $m = 0$ (+1 eigenvalue) is governed by the expectation value $P_0 = \frac{1}{2}(1 + \langle \psi_{\mathcal{X}^{(t)}} | P^{(t)} | \psi_{\mathcal{X}^{(t)}} \rangle)$. We take the inner product of the original state (indexed by $\boldsymbol{\beta} \subseteq \mathbf{b}^{(t)}$) against the shifted state (indexed by $\boldsymbol{\alpha} \subseteq \mathbf{b}^{(t)}$) as defined by Eq. 6 for $P^{(t)}$ decomposed to $(\omega, \mathbf{x}, \mathbf{z})$. Because physical destabilizers ($\bar{d}_i^\dagger = \bar{d}_i$) are orthogonal over the vacuum, the inner product evaluates to a Kronecker delta over $\mathbb{F}_2$:

$$\langle P^{(t)} \rangle = |\gamma^{(t)}|^2 \sum_{\boldsymbol{\beta} \subseteq \mathbf{b}^{(t)}} \sum_{\boldsymbol{\alpha} \subseteq \mathbf{b}^{(t)}} \left(a_{\boldsymbol{\beta}}^{(t)}\right)^* a_{\boldsymbol{\alpha}}^{(t)} \Phi(\boldsymbol{\alpha}; \omega, \mathbf{x}, \mathbf{z}, \mathbf{p}_X^{(t)}, \mathbf{p}_Z^{(t)}) \underbrace{\left\langle \mathbf{0}_{\mathcal{S}^{(t)}}(\mathbf{p}_Z^{(t)}) \middle| (\bar{\mathbf{d}}^{(t)})^{\boldsymbol{\beta} \oplus \boldsymbol{\alpha} \oplus \mathbf{x}} \middle| \mathbf{0}_{\mathcal{S}^{(t)}}(\mathbf{p}_Z^{(t)}) \right\rangle}_{\delta_{\boldsymbol{\beta} \oplus \boldsymbol{\alpha} \oplus \mathbf{x}, \mathbf{0}}}$$

This forces $\boldsymbol{\beta} = \boldsymbol{\alpha} \oplus \mathbf{x}$, collapsing the double sum. Because the bra sum restricts $\boldsymbol{\beta} \subseteq \mathbf{b}^{(t)}$, the surviving terms require the shifted branch to exist within the active mask:

$$\langle P^{(t)} \rangle = |\gamma^{(t)}|^2 \sum_{\substack{\boldsymbol{\alpha} \subseteq \mathbf{b}^{(t)} \\ (\boldsymbol{\alpha} \oplus \mathbf{x}) \subseteq \mathbf{b}^{(t)}}} \left(a_{\boldsymbol{\alpha} \oplus \mathbf{x}}^{(t)}\right)^* a_{\boldsymbol{\alpha}}^{(t)} \, \Phi(\boldsymbol{\alpha}; \omega, \mathbf{x}, \mathbf{z}, \mathbf{p}_X^{(t)}, \mathbf{p}_Z^{(t)})$$

The probablistic outcome determined by how the geometric shift $\mathbf{x}$ interacts with the support mask $\mathbf{b}^{(t)}$. We partition the extracted bitmasks: $\mathbf{x}_{\text{dorm}} = \mathbf{x} \wedge \neg \mathbf{b}^{(t)}$, $\mathbf{x}_{\text{act}} = \mathbf{x} \wedge \mathbf{b}^{(t)}$, and $\mathbf{z}_{\text{act}} = \mathbf{z} \wedge \mathbf{b}^{(t)}$. This isolates four physical cases.

Case 3a: Random Pivot / Vacuum Collapse ($\mathbf{x}_{\text{dorm}} \neq \mathbf{0}$)
The measurement anticommutes with at least one dormant stabilizer. Because the shift $\boldsymbol{\alpha} \mapsto \boldsymbol{\alpha} \oplus \mathbf{x}$ maps every active branch into a previously empty orthogonal subspace ($\boldsymbol{\alpha} \oplus \mathbf{x} \not\subseteq \mathbf{b}^{(t)}$), the expectation value evaluates to zero, and is a 50-50 outcome.

Explicit Update Rules ($\mathcal{X}^{(t)} \mapsto \mathcal{X}^{(t+1)}$):
We select a dormant source pivot $k_p \in \text{supp}(\mathbf{x}_{\text{dorm}})$. Let $\mathcal{T}_{\text{tmp}}$ and $\mathbf{p}_X^{\text{tmp}}, \mathbf{p}_Z^{\text{tmp}}$ represent the geometry and algebraic parities after iteratively applying $\Lambda_{\text{rowsum}}(i, k_p)$ for all $i \in \text{supp}(\mathbf{x}) \backslash \{k_p\}$. The complex array a remains untouched via the Zero-Cost Dormant Property, while the runtime tracking parities shift accordingly.

We re-extract the observable as $(\omega', \mathbf{x}', \mathbf{z}') = \Gamma(P^{(t)}, \mathcal{T}_{\text{tmp}})$. Because the prerequisite rowsums

clear all X-support except the pivot, $\mathbf{x}'$ is guaranteed to be $\mathbf{e}_{k_p}$.

$$
\begin{aligned}
P_0 &= 0.5 && \text{(Absolute expectation)} \\
m &= \begin{cases} 0 & \text{if } \xi^{(t)} < P_0 \\ 1 & \text{if } \xi^{(t)} \geq P_0 \end{cases} && \text{(Sample outcome } m\text{)} \\
\mathcal{M}^{(t+1)} &= \mathcal{M}^{(t)} \| \langle m \rangle && \text{(Append outcome to sequence)} \\
\mathcal{T}^{(t+1)} &= \Lambda_{\text{pivot}}(k_p, \omega', \mathbf{z}', \mathcal{T}_{\text{tmp}}) && \text{(AG Pivot)} \\
\mathbf{b}^{(t+1)} &= \mathbf{b}^{(t)} && \text{(Dimensionality preserved)} \\
\gamma^{(t+1)} &= \gamma^{(t)} && \text{(Invariant magnitude)} \\
p_{Z,j}^{(t+1)} &= p_{Z,j}^{\text{tmp}} && \forall j \neq k_p \\
p_{X,j}^{(t+1)} &= p_{X,j}^{\text{tmp}} \oplus (z'_j \cdot p_{Z,k_p}^{\text{tmp}}) && \forall j \neq k_p \\
p_{X,k_p}^{(t+1)} &= p_{Z,k_p}^{\text{tmp}}, \quad p_{Z,k_p}^{(t+1)} = m && \text{(Anchor new vacuum)} \\
a_{\boldsymbol{\alpha}}^{(t+1)} &= a_{\boldsymbol{\alpha}}^{(t)} \quad \forall \boldsymbol{\alpha} \subseteq \mathbf{b}^{(t+1)} && \text{(Identity map)}
\end{aligned}
$$

Case 3b: Interfering Shrink ($\mathbf{x}_{\text{dorm}} = \mathbf{0}$ and $\mathbf{x}_{\text{act}} \neq \mathbf{0}$)
The measurement contains only active destabilizers, meaning it anticommutes with the corresponding active stabilizers. This means ($\boldsymbol{\alpha}$ interferes with $\boldsymbol{\alpha} \oplus \mathbf{x}$). The projection halves the active dimension as the specific paired branches fold over each other. Because folding the paired branches yields identical squared magnitudes for both $\boldsymbol{\alpha}$ and $\boldsymbol{\alpha} \oplus \mathbf{e}_{k_p}$, summing over half the space requires a multiplication factor of $2 \times \frac{1}{4} = \frac{1}{2}$ (where the 1/4 comes from squaring the projector $\frac{1}{2}(I \pm P)$).

Explicit Update Rules ($\mathcal{X}^{(t)} \mapsto \mathcal{X}^{(t+1)}$):
We select an active source pivot $k_p \in \text{supp}(\mathbf{x}_{\text{act}})$. Let $\mathcal{T}_{\text{tmp}}$ be the geometry after iteratively applying $\Lambda_{\text{rowsum}}(i, k_p)$ for all $i \in \text{supp}(\mathbf{x}_{\text{act}}) \setminus \{k_p\}$. Because k_p is an active source, the tuple undergoes the Amplitude Reindexing Map. Let a^{tmp} and $\mathbf{p}_X^{\text{tmp}}, \mathbf{p}_Z^{\text{tmp}}$ be the intermediate state after systematically applying Eq. 9. We re-extract $(\omega', \mathbf{x}', \mathbf{z}') = \Gamma(P^{(t)}, \mathcal{T}_{\text{tmp}})$, aligning the observable such that $\mathbf{x}' = \mathbf{e}_{k_p}$, and evaluate the dynamic phase $\Phi'(\boldsymbol{\alpha}) := \omega'(-1)^{p_{X,k_p}^{\text{tmp}} + \mathbf{z}' \cdot \mathbf{p}_Z^{\text{tmp}} + \mathbf{z}' \cdot \boldsymbol{\alpha}} = \Phi(\boldsymbol{\alpha}; \omega', \mathbf{e}_{k_p}, \mathbf{z}', \mathbf{p}_X^{\text{tmp}}, \mathbf{p}_Z^{\text{tmp}})$.

The transition rules are then:

$$
\begin{aligned}
\mathbf{b}^{(t+1)} &= \mathbf{b}^{(t)} \setminus \{k_p\} && \text{(Dimension halves)} \\
p_0 &= \sum_{\boldsymbol{\alpha} \subseteq \mathbf{b}^{(t+1)}} \frac{1}{2} \left| a^{\text{tmp}}_{\boldsymbol{\alpha}} + \Phi'(\boldsymbol{\alpha} \oplus \mathbf{e}_{k_p}) \cdot a^{\text{tmp}}_{\boldsymbol{\alpha} \oplus \mathbf{e}_{k_p}} \right|^2 && \text{(Unnormalized structural weight)} \\
P_0 &= p_0 \cdot |\gamma^{(t)}|^2 && \text{(Absolute expectation)} \\
m &= \begin{cases} 0 & \text{if } \xi^{(t)} < P_0 \\ 1 & \text{if } \xi^{(t)} \geq P_0 \end{cases} && \text{(Sample outcome } m\text{; let } P_m \text{ be its prob.)} \\
\mathcal{M}^{(t+1)} &= \mathcal{M}^{(t)} \| \langle m \rangle && \text{(Append outcome to sequence)} \\
\mathcal{T}^{(t+1)} &= \Lambda_{\text{pivot}}(k_p, \omega', \mathbf{z}', \mathcal{T}_{\text{tmp}}) && \text{(Demote } k_p \text{ to vacuum)} \\
\gamma^{(t+1)} &= \frac{\gamma^{(t)}}{\sqrt{2P_m}} && \text{(Deferred Normalization)} \\
p^{(t+1)}_{Z,j} &= p^{\text{tmp}}_{Z,j} && \forall j \neq k_p \\
p^{(t+1)}_{X,j} &= p^{\text{tmp}}_{X,j} \oplus (z'_j \cdot p^{\text{tmp}}_{Z,k_p}) && \forall j \neq k_p \\
p^{(t+1)}_{X,k_p} &= p^{\text{tmp}}_{Z,k_p}, \quad p^{(t+1)}_{Z,k_p} = m && \text{(Anchor new vacuum)} \\
a^{(t+1)}_{\boldsymbol{\alpha}} &= a^{\text{tmp}}_{\boldsymbol{\alpha}} + (-1)^m \cdot \Phi'(\boldsymbol{\alpha} \oplus \mathbf{e}_{k_p}) \cdot a^{\text{tmp}}_{\boldsymbol{\alpha} \oplus \mathbf{e}_{k_p}} && \forall \boldsymbol{\alpha} \subseteq \mathbf{b}^{(t+1)} \quad \text{(Division-free addition)}
\end{aligned}
$$

Case 3c: Diagonal Shrink ($\mathbf{x} = \mathbf{0}$ and $\mathbf{z}_{\text{act}} \neq \mathbf{0}$)
The measurement evaluates diagonally on the active branches. The coordinate shift is trivial ($\boldsymbol{\alpha} \oplus \mathbf{0} = \boldsymbol{\alpha}$). The structural expectation evaluates the physical magnitude matching the deterministic phase constraint, filtering the unnormalized branches without any mixing.

Explicit Update Rules ($\mathcal{X}^{(t)} \mapsto \mathcal{X}^{(t+1)}$):
We select an active target pivot $k_p \in \text{supp}(\mathbf{z}_{\text{act}})$. Let $\mathcal{T}_{\text{tmp}}$ evaluate the geometry after applying target-swapped rowsums $\Lambda_{\text{rowsum}}(k_p, i)$ for all $i \in \text{supp}(\mathbf{z}_{\text{act}}) \setminus \{k_p\}$. Because i is an active source, the tuple applies the Amplitude Reindexing Map, yielding a^{tmp} and $\mathbf{p}^{\text{tmp}}_X, \mathbf{p}^{\text{tmp}}_Z$. We re-extract $(\omega', \mathbf{0}, \mathbf{z}') = \Gamma(P^{(t)}, \mathcal{T}_{\text{tmp}})$ and evaluate the diagonal phase $\Phi'(\boldsymbol{\alpha}) := \omega'(-1)^{\mathbf{z}' \cdot \mathbf{p}^{\text{tmp}}_Z + \boldsymbol{\alpha}_{k_p}}$.

$$
\begin{aligned}
\mathbf{b}^{(t+1)} &= \mathbf{b}^{(t)} \setminus \{k_p\} && \text{(Dimension halves)} \\
p_0 &= \sum_{\substack{\boldsymbol{\alpha} \subseteq \mathbf{b}^{(t)} \\ \Phi'(\boldsymbol{\alpha}) = +1}} \left| a^{\text{tmp}}_{\boldsymbol{\alpha}} \right|^2 && \text{(Unnormalized structural weight)} \\
P_0 &= p_0 \cdot |\gamma^{(t)}|^2 && \text{(Absolute expectation)} \\
m &= \begin{cases} 0 & \text{if } \xi^{(t)} < P_0 \\ 1 & \text{if } \xi^{(t)} \geq P_0 \end{cases} && \text{(Sample outcome } m\text{; let } P_m \text{ be its prob.)} \\
\mathcal{M}^{(t+1)} &= \mathcal{M}^{(t)} \| \langle m \rangle && \text{(Append outcome to sequence)} \\
\mathcal{T}^{(t+1)} &= \mathcal{T}_{\text{tmp}} && \text{(Gauge Transformation)}
\end{aligned}
$$

Let $v \in \{0, 1\}$ denote the unique Boolean slice across the k_p-th axis satisfying the evaluation

constraint, ensuring $\Phi'(\boldsymbol{\alpha} \text{ where } \boldsymbol{\alpha}_{k_p} = v) = (-1)^m$. We filter the valid remainder:

$$
\begin{aligned}
\gamma^{(t+1)} &= \frac{\gamma^{(t)}}{\sqrt{P_m}} \cdot (-1)^{p^{\text{tmp}}_{X,k_p} \cdot v} && \text{(Deferred Normalization \& Phase)} \\
p^{(t+1)}_{X,j} &= p^{\text{tmp}}_{X,j}, \quad p^{(t+1)}_{Z,j} = p^{\text{tmp}}_{Z,j} && \forall j \neq k_p \\
p^{(t+1)}_{X,k_p} &= p^{\text{tmp}}_{X,k_p}, \quad p^{(t+1)}_{Z,k_p} = p^{\text{tmp}}_{Z,k_p} \oplus v && \text{(Anchor runtime parity)} \\
a^{(t+1)}_{\boldsymbol{\alpha}} &= a^{\text{tmp}}_{\boldsymbol{\alpha} \oplus (v \cdot \mathbf{e}_{k_p})} && \forall \boldsymbol{\alpha} \subseteq \mathbf{b}^{(t+1)} \quad \text{(Division-free slice filter)}
\end{aligned}
$$

Case 3d: Deterministic Filter ($\mathbf{x} = \mathbf{0}$ and $\mathbf{z}_{\mathbf{act}} = \mathbf{0}$)
The measurement evaluates to a deterministic constant against the physical vacuum state. The geometry and specific probability states pass invariant.
Explicit Update Rules ($\mathcal{X}^{(t)} \mapsto \mathcal{X}^{(t+1)}$):

$$
\begin{aligned}
m &= \begin{cases} 0 & \text{if } \omega(-1)^{\mathbf{z} \cdot \mathbf{p}^{(t)}_Z} = +1 \\ 1 & \text{if } \omega(-1)^{\mathbf{z} \cdot \mathbf{p}^{(t)}_Z} = -1 \end{cases} && \text{(Deterministic evaluation)} \\
\mathcal{M}^{(t+1)} &= \mathcal{M}^{(t)} \| \langle m \rangle && \text{(Append outcome to sequence)} \\
\mathcal{T}^{(t+1)} &= \mathcal{T}^{(t)} \\
\mathbf{b}^{(t+1)} &= \mathbf{b}^{(t)} \\
\gamma^{(t+1)} &= \gamma^{(t)} \\
\mathbf{p}^{(t+1)}_X &= \mathbf{p}^{(t)}_X, \quad \mathbf{p}^{(t+1)}_Z = \mathbf{p}^{(t)}_Z \\
a^{(t+1)}_{\boldsymbol{\alpha}} &= a^{(t)}_{\boldsymbol{\alpha}} \quad \forall \boldsymbol{\alpha} \subseteq \mathbf{b}^{(t+1)} && \text{(Identity map)}
\end{aligned}
$$

4.4 Conditional Pauli Operations

We unify stochastic Pauli noise and classical conditional feedback by defining a generic conditional Pauli operation $P^{(t)}(b)$. The physical operator $P^{(t)}$ applies if a Boolean trigger $b \in \{0, 1\}$ evaluates to 1. Specifically:

- **Stochastic Pauli Noise:** Conditioned on an isolated probability rate $p^{(t)}_E$ queried against the random variable $\xi^{(t)}$: $b = \mathbf{1}_{\xi^{(t)} < p^{(t)}_E}$.
- **Classical Feedback:** Conditioned on a prior classical measurement, the random stream is bypassed and b evaluates to the k-th element of the classical memory sequence:

$$
b = m^{(k)} \quad (\text{where } m^{(k)} \in \mathcal{M}^{(t)}) \tag{11}
$$

Derivation: Assuming the condition evaluates to true ($b = 1$), applying $P^{(t)}$ to the state evaluates via the Pauli action derived in Eq. 6:

$$
P^{(t)} \left| \psi_{\mathcal{X}^{(t)}} \right\rangle = \gamma^{(t)} \sum_{\boldsymbol{\alpha} \subseteq \mathbf{b}^{(t)}} a^{(t)}_{\boldsymbol{\alpha}} \omega (-1)^{\mathbf{x} \cdot \mathbf{p}^{(t)}_X + \mathbf{z} \cdot \mathbf{p}^{(t)}_Z + \mathbf{z} \cdot \boldsymbol{\alpha}} \left((\bar{\mathbf{d}}^{(t)})^{\boldsymbol{\alpha} \oplus \mathbf{x}} \left| \mathbf{0}_{\mathcal{S}^{(t)}} (\mathbf{p}^{(t)}_Z) \right\rangle \right)
$$

We absorb this coordinate transform into the tracking parities without mutating the complex array $a^{(t)}$. Let the updated parities be $\mathbf{p}^{(t+1)}_Z = \mathbf{p}^{(t)}_Z \oplus \mathbf{x}$ and $\mathbf{p}^{(t+1)}_X = \mathbf{p}^{(t)}_X \oplus \mathbf{z}$.

First, shifting the runtime Z-parity redefines the physical vacuum state. Because applying destabilizers flips the corresponding stabilizer eigenvalues (e.g s_i on $d_i |0_i\rangle$ now has a -1 eigenvalue), the geometric shift maps the vacuum via the un-signed generators: $\left|\mathbf{0}_{\mathcal{S}^{(t+1)}}(\mathbf{p}_Z^{(t+1)})\right\rangle = (\mathbf{d}^{(t)})^{\mathbf{x}} \left|\mathbf{0}_{\mathcal{S}^{(t)}}(\mathbf{p}_Z^{(t)})\right\rangle$. Substituting the physical identity $(\bar{\mathbf{d}}^{(t)})^{\mathbf{x}} = (-1)^{\mathbf{x}\cdot\mathbf{p}_X^{(t)}}(\mathbf{d}^{(t)})^{\mathbf{x}}$ isolates the new vacuum:

$$(\bar{\mathbf{d}}^{(t)})^{\mathbf{x}} \left|\mathbf{0}_{\mathcal{S}^{(t)}}(\mathbf{p}_Z^{(t)})\right\rangle = (-1)^{\mathbf{x}\cdot\mathbf{p}_X^{(t)}} \left|\mathbf{0}_{\mathcal{S}^{(t+1)}}(\mathbf{p}_Z^{(t+1)})\right\rangle$$

Second, shifting the X-parity updates the physical destabilizers via $\bar{d}_i^{(t+1)} = (-1)^{z_i}\bar{d}_i^{(t)}$, which establishes the branch relationship $(\bar{\mathbf{d}}^{(t)})^{\boldsymbol{\alpha}} = (-1)^{\mathbf{z}\cdot\boldsymbol{\alpha}}(\bar{\mathbf{d}}^{(t+1)})^{\boldsymbol{\alpha}}$.

Because physical destabilizers mutually commute, $(\bar{\mathbf{d}}^{(t)})^{\boldsymbol{\alpha}\oplus\mathbf{x}} = (\bar{\mathbf{d}}^{(t)})^{\boldsymbol{\alpha}}(\bar{\mathbf{d}}^{(t)})^{\mathbf{x}}$. Substituting these identities directly into the Pauli action equation yields:

$$\begin{aligned}
P^{(t)} |\psi_{\mathcal{X}^{(t)}}\rangle &= \gamma^{(t)} \sum_{\boldsymbol{\alpha}\subseteq\mathbf{b}^{(t)}} a_{\boldsymbol{\alpha}}^{(t)} \omega(-1)^{\mathbf{x}\cdot\mathbf{p}_X^{(t)}+\mathbf{z}\cdot\mathbf{p}_Z^{(t)}+\mathbf{z}\cdot\boldsymbol{\alpha}} \Big((\bar{\mathbf{d}}^{(t)})^{\boldsymbol{\alpha}} \underbrace{(-1)^{\mathbf{x}\cdot\mathbf{p}_X^{(t)}} \left|\mathbf{0}_{\mathcal{S}^{(t+1)}}(\mathbf{p}_Z^{(t+1)})\right\rangle}_{\text{shifted vacuum}} \Big) \\
&= \gamma^{(t)} \sum_{\boldsymbol{\alpha}\subseteq\mathbf{b}^{(t)}} a_{\boldsymbol{\alpha}}^{(t)} \omega(-1)^{\mathbf{z}\cdot\mathbf{p}_Z^{(t)}+\mathbf{z}\cdot\boldsymbol{\alpha}} \Big(\underbrace{(-1)^{\mathbf{z}\cdot\boldsymbol{\alpha}}(\bar{\mathbf{d}}^{(t+1)})^{\boldsymbol{\alpha}}}_{(\bar{\mathbf{d}}^{(t)})^{\boldsymbol{\alpha}}} \left|\mathbf{0}_{\mathcal{S}^{(t+1)}}(\mathbf{p}_Z^{(t+1)})\right\rangle \Big) \\
&= \left[\gamma^{(t)}\omega(-1)^{\mathbf{z}\cdot\mathbf{p}_Z^{(t)}}\right] \sum_{\boldsymbol{\alpha}\subseteq\mathbf{b}^{(t)}} a_{\boldsymbol{\alpha}}^{(t)} (\bar{\mathbf{d}}^{(t+1)})^{\boldsymbol{\alpha}} \left|\mathbf{0}_{\mathcal{S}^{(t+1)}}(\mathbf{p}_Z^{(t+1)})\right\rangle
\end{aligned}$$

Because $(-1)^{2(\mathbf{x}\cdot\mathbf{p}_X^{(t)})} = 1$ and $(-1)^{2(\mathbf{z}\cdot\boldsymbol{\alpha})} = 1$, the X-parity phase and localized array dynamic phase cancel over $\mathbb{F}_2$. This leaves the complex array $a^{(t)}$ invariant, collapsing the physical coordinate transform into a scalar update on $\gamma^{(t)}$.

To integrate the case where the condition evaluates to false ($b = 0$), we patch the Boolean indicator b directly into the algebraic updates.

Explicit Update Rules ($\mathcal{X}^{(t)}, \xi^{(t)} \mapsto \mathcal{X}^{(t+1)}$):
By expressing the Boolean trigger via an algebraic indicator polynomial, the conditional unnormalized structural updates apply exclusively via the tracker γ:

$$\begin{aligned}
\gamma^{(t+1)} &= \gamma^{(t)} \cdot \left[(1-b) + b\cdot\omega(-1)^{\mathbf{z}\cdot\mathbf{p}_Z^{(t)}}\right] \\
\mathbf{p}_Z^{(t+1)} &= \mathbf{p}_Z^{(t)} \oplus (b\cdot\mathbf{x}) \\
\mathbf{p}_X^{(t+1)} &= \mathbf{p}_X^{(t)} \oplus (b\cdot\mathbf{z})
\end{aligned}$$

5 The UCC Architecture

A direct consequence of the partition established in Section 3 is **Symplectic Determinism**. The evolution of the geometric state $(\mathcal{T}^{(t)}, \mathbf{b}^{(t)})$ depends exclusively on the sequence of operations applied. It is independent of the random outcomes m of measurements, the probabilistic triggers of stochastic noise, and the continuous complex values of the unnormalized amplitudes $a^{(t)}$.

Because the geometric frame's symplectic trajectory and its structural sign changes are predetermined by the circuit's topological structure, $\mathcal{T}^{(t)}$ can be decoupled from the probabilistic execution thread. We map this separation into a concrete software architecture—the Universal Compiler Collection (UCC)—which divides execution between an offline Ahead-Of-Time (AOT) compiler and an online runtime virtual machine (the Schrodinger VM), summarized in Table 1.

Circuit Operation $O^{(t)}$	**AOT Compiler (Offline)** Updates $\mathcal{T}, \mathbf{b}$	**VM Payload** Emitted by AOT	**Schrodinger VM (Online)** Updates $\mathbf{p}_{X,Z}, \gamma, a, \mathcal{M}$
Clifford (U)	$\mathcal{T} \leftarrow \Lambda_C(U, \mathcal{T})$ $\theta_{\text{global}} \leftarrow \theta_{\text{global}} + \theta_U$	*(Absorbed into Header)* θ_{global}	*(Applied at VM startup)* $\gamma^{(0)} \leftarrow e^{i\theta_{\text{global}}}$
Non-Clifford	Rowsums $\rightarrow \mathcal{T}_{\text{tmp}}$ $\mathbf{b} \leftarrow \mathbf{b} \vee \mathbf{x}'_{\text{dorm}}$	`OP_MIX` $\omega', \mathbf{x}', \mathbf{z}'$, weight c	Scale magnitude γ; Mix branches a
Meas 3a (Random Pivot)	Rowsums $\rightarrow \mathcal{T}_{\text{tmp}}$ $\mathcal{T} \leftarrow \Lambda_{\text{pivot}}$	`OP_MEAS_RANDOM` $\omega', k_p, \mathbf{z}'$	Sample absolute P_0; Anchor parities; Record m
Meas 3b (Interfere)	Rowsums $\rightarrow \mathcal{T}_{\text{tmp}}$ $\mathcal{T} \leftarrow \Lambda_{\text{pivot}}$; $\mathbf{b} \leftarrow \mathbf{b} \setminus \{k_p\}$	`OP_MEAS_INTERFERE` $\omega', k_p, \mathbf{z}'$	Calc P_0; Sample m; Fold a; Anchor parities; Scale γ
Meas 3c (Diagonal)	Target-swapped Rowsums $\mathbf{b} \leftarrow \mathbf{b} \setminus \{k_p\}$	`OP_MEAS_DIAGONAL` $\omega', k_p, \mathbf{z}'$	Calc P_0; Sample m; Filter a; Anchor parities; Scale γ
Meas 3d (Static Filter)	*Identity Map*	`OP_MEAS_STATIC` $\omega, \mathbf{z}$	Eval static m; Record m
Stoch. Noise	*Identity Map*	`OP_NOISE` $\omega, \mathbf{x}, \mathbf{z}, p_E$	Eval rand ξ; Update γ; $\mathbf{p} \leftarrow \mathbf{p} \oplus \mathbf{x}, \mathbf{z}$
Class. Feedback	*Identity Map*	`OP_FEEDBACK` $\omega, \mathbf{x}, \mathbf{z}, k$	Eval trigger $m^{(k)}$; Update γ; $\mathbf{p} \leftarrow \mathbf{p} \oplus \mathbf{x}, \mathbf{z}$

Table 1: The Compiler-VM API bridging the geometric and probabilistic domains. The offline compiler performs heavy $\mathcal{O}(n^2)$ tableau manipulations, emitting only lightweight vector bitmasks $(\mathbf{x}', \mathbf{z}')$ and scalars (ω', k_p, θ_U). The probabilistic runtime uses these payloads to orchestrate complex branch interference utilizing $\mathcal{O}(1)$ division-free array logic due to the integration of Deferred Normalization. Active-source rowsums also emit `OP_REINDEX` instructions (omitted for brevity) to safely reorder the vector array a.

This API guarantees that the $\mathcal{O}(n^2)$ tableau operations are isolated to the left column. The runtime probabilistic execution thread (the right column) executes exclusively over minimal arrays, utilizing division-free logic due to the integration of Deferred Normalization. The UCC architecture implements this partition in four distinct stages: the Front-End, the Middle-End Optimizer, the Compiler Back-End, and the Schrodinger VM.

5.1 The Algebraic Equivalence of the HIR

To enable global classical optimization without the computational overhead of propagating a local geometric frame, the compiler maps the input circuit into the Heisenberg Intermediate Representation (HIR). We establish the HIR as a parallel construct by proving that tracking rewound operations is equivalent to evaluating the original circuit.

Let a quantum circuit be an alternating sequence of deterministic Clifford operations and active operations (such as non-Clifford rotations, Pauli measurements, conditional Pauli noise, or classical feedback). Because active operations are not unitary (e.g., projective measurements), we represent the circuit as a composition of quantum maps (or channels). For any contiguous run of uninter-

rupted Clifford gates, we collapse them into a single equivalent Clifford map $\mathcal{C}_k(\rho) = C_k \rho C_k^\dagger$. While collapsing these blocks abstracts away localized hardware routing information, it preserves the exact logical operation. Let $\mathcal{E}_k$ denote the k-th active operation map. The full sequence evaluates as:

$$\mathcal{O}_{\text{total}} = \mathcal{C}_M \circ \mathcal{E}_M \circ \mathcal{C}_{M-1} \circ \mathcal{E}_{M-1} \cdots \circ \mathcal{C}_1 \circ \mathcal{E}_1 \circ \mathcal{C}_0 \tag{12}$$

We define the cumulative Clifford map $\mathcal{U}_k$ as the total deterministic geometric transformation applied from the beginning of the circuit up to step k:

$$\mathcal{U}_k = \mathcal{C}_k \circ \mathcal{C}_{k-1} \cdots \circ \mathcal{C}_0 \tag{13}$$

It follows that $\mathcal{U}_0 = \mathcal{C}_0$. Because Clifford maps are automorphisms generated by unitary matrices, they are invertible. Any local Clifford block at step $k > 0$ can be factored exactly as the transformation between two cumulative frames: $\mathcal{C}_k = \mathcal{U}_k \circ \mathcal{U}_{k-1}^{-1}$. Substituting this identity back into the circuit yields a telescoping sequence:

$$\mathcal{O}_{\text{total}} = \mathcal{U}_M \circ (\mathcal{U}_{M-1}^{-1} \circ \mathcal{E}_M \circ \mathcal{U}_{M-1}) \circ \cdots \circ (\mathcal{U}_0^{-1} \circ \mathcal{E}_1 \circ \mathcal{U}_0) \tag{14}$$

Defining the Heisenberg-rewound active operation map as $\tilde{\mathcal{E}}_k := \mathcal{U}_{k-1}^{-1} \circ \mathcal{E}_k \circ \mathcal{U}_{k-1}$, the total circuit operator simplifies to a flat sequence:

$$\mathcal{O}_{\text{total}} = \mathcal{U}_M \circ \tilde{\mathcal{E}}_M \circ \tilde{\mathcal{E}}_{M-1} \cdots \circ \tilde{\mathcal{E}}_1 \tag{15}$$

Crucially, every active map $\mathcal{E}_k$ in our circuit model is structured around a physical Pauli generator P_k (e.g., projecting onto P_k, rotating around P_k, or applying noise P_k). Conjugating the entire map by a cumulative Clifford frame U distributes to the underlying generator. For example, the operator for a projective measurement $\Pi_k = \frac{1}{2}(I \pm P_k)$ maps to:

$$U^\dagger \left[\frac{1}{2}(I \pm P_k)\right] U = \frac{1}{2}\big(I \pm U^\dagger P_k U\big) = \frac{1}{2}(I \pm \tilde{P}_k) \tag{16}$$

Similarly, a unitary rotation $e^{i\theta P_k}$ maps to $e^{i\theta \tilde{P}_k}$. The Clifford group normalizes the Pauli group, guaranteeing that the transformed generator $\tilde{P}_k$ is mapped to another valid Pauli string.

This collapse demonstrates a 1-to-1 mapping: executing local active operations dynamically amidst a shifting geometric frame ($\mathcal{E}_k$) is identical to composing a flat sequence of rewound operations ($\tilde{\mathcal{E}}_k$) parameterized by $t = 0$ Paulis, evaluated within the static $t = 0$ reference frame, followed by a single global geometric coordinate transformation $\mathcal{U}_M$ at the very end. This derivation relies on map identities and is independent of the initial state.

5.2 The Front-End

The Front-End compiler performs this rewinding mapping. It tracks the cumulative frame U_k via stabilizer tableaus, absorbs the intermediate C_k blocks, rewinds each active operation to find $\tilde{P}_k$, and evaluates the Pauli Extraction Map $\Gamma(\tilde{P}_k, \mathcal{T}^{(0)})$ against the initial computational tableau $\mathcal{T}^{(0)}$ to isolate the bitwise masks $(\omega, \mathbf{x}, \mathbf{z})$.

This sequence of extracted $t = 0$ masks forms the HIR. Because deterministic Cliffords are geometrically absorbed by the tracking of U_k, their topological structure vanishes from the representation. However, because symplectic conjugation mathematically drops the continuous global phase (i.e., $U^\dagger P U$ is identical for U and $e^{i\theta}U$), the Front-End explicitly extracts the continuous phase θ_U of each absorbed Clifford and accumulates it into a single, circuit-wide scalar θ_{global}. The HIR is therefore a flat sequence of abstract active operations parameterized globally by this single scalar. Each operation type stores its extracted $t = 0$ Pauli string alongside its semantic parameters:

- `HIR_NON_CLIFFORD`: Represents a parameterized rotation expanded as a sum of Pauli terms. Stores the absolute Pauli operator and the relative complex weight c of the target branch.
- `HIR_MEASURE`: Represents a projective state collapse. Stores the absolute Pauli observable and a target integer index for the classical memory sequence $\mathcal{M}$.
- `HIR_NOISE`: Represents a stochastic Pauli error channel. Stores the absolute Pauli error operator and the continuous physical probability p_E of the error occurring.
- `HIR_FEEDBACK`: Represents a classically controlled Pauli operation. Stores the absolute Pauli operator and the source integer index k pointing to a previously recorded outcome $m^{(k)} \in \mathcal{M}$.

5.3 The Middle-End Optimizer

By expressing all operations in a unified $t = 0$ coordinate system, the Middle-End Optimizer can evaluate circuit relationships globally without requiring iterative geometric frame updates.

To a skeptical reader, it might seem perilous to reorder or fuse active operations A and B that occur at different times, separated by intermediate operations, and within distinct geometric Clifford frames in the original circuit. However, we can justify this by proving that physical commutation is invariant beneath a geometric coordinate transformation, and by leveraging the algebraic properties of quantum map composition.

First, suppose P_A and P_B are physical Pauli generators evaluated in a localized cumulative frame U. Their rewound $t = 0$ equivalents are $\tilde{P}_A = U^\dagger P_A U$ and $\tilde{P}_B = U^\dagger P_B U$. Expanding the commutator of these rewound operators yields:

$$\begin{aligned}
[\tilde{P}_A, \tilde{P}_B] &= \tilde{P}_A \tilde{P}_B - \tilde{P}_B \tilde{P}_A \\
&= (U^\dagger P_A U)(U^\dagger P_B U) - (U^\dagger P_B U)(U^\dagger P_A U) \\
&= U^\dagger (P_A P_B) U - U^\dagger (P_B P_A) U \\
&= U^\dagger [P_A, P_B] U
\end{aligned}$$

Because U is a unitary matrix, $[\tilde{P}_A, \tilde{P}_B] = 0$ if and only if $[P_A, P_B] = 0$. This guarantees that operations commute in their true forward-moving runtime frame if and only if their extracted $t = 0$ Pauli generators commute.

Second, this property allows the compiler to bridge the temporal distance between operations separated by *other* active operations. Because Section 5.1 formalizes the circuit into a flat sequence of composed maps (e.g., $\cdots \circ \tilde{\mathcal{E}}_B \circ \tilde{\mathcal{E}}_X \circ \tilde{\mathcal{E}}_A \circ \ldots$), the compiler can reorder them pairwise. Because the rewound maps $\tilde{\mathcal{E}}$ rely on their $t = 0$ Pauli generators, standard quantum mechanics dictates that adjacent maps commute ($\tilde{\mathcal{E}}_X \circ \tilde{\mathcal{E}}_A = \tilde{\mathcal{E}}_A \circ \tilde{\mathcal{E}}_X$) if and only if their underlying $t = 0$ Paulis commute.

To evaluate this, the optimizer computes the symplectic inner product over $\mathbb{F}_2$ between the $t = 0$ bitmasks of operation A and an intervening operation X:

$$(\mathbf{x}_A \cdot \mathbf{z}_X) \oplus (\mathbf{z}_A \cdot \mathbf{x}_X) \equiv 0 \pmod 2 \tag{17}$$

This single $\mathcal{O}(1)$ bitwise check determines commutativity. If it evaluates to zero, the compiler swaps their execution order, allowing operation A to "slide" past X. Through successive commuting swaps (similar to a bubble sort), distant operations can become adjacent. Once adjacent, the optimizer compresses the circuit using the following rules:

- **Algebraic Fusion:** If adjacent `HIR_NON_CLIFFORD` operations target the exact same topological Pauli axis ($\mathbf{x}_A = \mathbf{x}_B$ and $\mathbf{z}_A = \mathbf{z}_B$), they are fused and their relative complex weights combined. This can cancel the branch entirely or promote it to a geometric Clifford operation (e.g., $T \cdot T = e^{i\pi/4}S$). If promoted to a Clifford, the node is absorbed: the compiler explicitly conjugates all subsequent HIR nodes by this new Clifford to update their $\mathbf{x}$ and $\mathbf{z}$ bitmasks, and extracts the resulting mathematically generated scalar phase (e.g., $\pi/4$) to add to the global accumulator θ_{global}.

- **Stochastic Barriers:** To preserve the correct chronological evaluation of the physical state equations derived in Section 3, the `HIR_MEASURE`, `HIR_NOISE`, and `HIR_FEEDBACK` nodes act as strict physical barriers. The optimizer will not reorder operations across these barriers if they anticommute, ensuring the physical observable eigenvalues and dynamic phase trajectories remain intact.

5.4 The Compiler Back-End (Code Generation)

Once the HIR is optimized in the global $t = 0$ frame, the Compiler Back-End translates it back into the local, forward-moving geometric frame $\mathcal{T}^{(t)}$ required by the runtime state evolution rules.

The Back-End steps sequentially through the HIR, simulating the geometric state offline. For each operation, it projects the $t = 0$ operator into the current intermediate frame $\mathcal{T}^{(t)}$. It performs the $\mathcal{O}(n^2)$ Symplectic Rowsums and Aaronson-Gottesman (AG) Pivots to resolve the geometric topology and extracts the physical, aligned payloads $(\omega', \mathbf{x}', \mathbf{z}', k_p)$ required by the VM.

Memory Pre-allocation: Simultaneously, the Back-End tracks the active support mask $\mathbf{b}^{(t)}$ step-by-step. Because dimensionality expansion occurs deterministically, the compiler computes the peak active dimension reached during the entire circuit: $k_{\max} = \max_t |\mathbf{b}^{(t)}|$. Knowing this maximum active rank bounds the required runtime memory, allowing the Back-End to embed this limit $k_{\max}$—alongside the final global phase accumulator θ_{global}—into the static header of the bytecode. This eliminates dynamic memory allocation during execution and avoids emitting $\mathcal{O}(G_{\text{clifford}})$ individual phase instructions into the payload.

5.5 The Schrodinger VM and Opcodes

The Schrodinger VM is a minimal execution environment that exclusively manages the probabilistic state variables: the classical memory sequence $\mathcal{M}$, the global phase tracking variable γ, the runtime parity trackers $\mathbf{p}_X, \mathbf{p}_Z$, and the complex amplitude array a.

At startup, the VM reads $k_{\max}$ and the global phase accumulator θ_{global} from the compiler payload header. It statically allocates the complex array a to $2^{k_{\max}}$ entries. The array is initialized with zeros, setting $a_{\mathbf{0}} = 1$. It also initializes the global magnitude tracker using the accumulated phase: $\gamma^{(0)} = e^{i\theta_{\text{global}}}$. The VM then iterates through the opcodes emitted by the Back-End, executing the physical updates defined by the unified state evolution equations in Section 3:

- `OP_REINDEX`:
  Payload: Target index i, Source index j.
  Action: Emitted when the Back-End performs an active-source rowsum offline. Instructs the VM to physically swap and reindex amplitudes in the array a to satisfy the Amplitude Reindexing Map (Eq. 9), assigning $a_{\boldsymbol{\beta}} \leftarrow a_{\boldsymbol{\beta} \oplus (\boldsymbol{\beta}_j \cdot \mathbf{e}_i)}$. The zero-cost dormant property guarantees this opcode is omitted for dormant sources.

- `OP_MIX`:
  Payload: Phase ω', masks $\mathbf{x}', \mathbf{z}'$, relative complex weight c.

Action: Evaluates the aligned dynamic phase $\Phi'(\boldsymbol{\alpha})$ against the current parities $\mathbf{p}_X, \mathbf{p}_Z$. Linearly mixes paired amplitudes $a_{\boldsymbol{\alpha}}$ and $a_{\boldsymbol{\alpha}\oplus\mathbf{x}'}$ utilizing the weight c. Updates the global magnitude tracker γ. Because the array is already allocated to the maximum dimension $2^{k_{\max}}$, structural expansion happens by updating the local active bounds and mixing with the pre-allocated zeros.

- `OP_MEAS_RANDOM`**:** (Maps to Case 3a)
  Payload: Phase ω', pivot index k_p, aligned mask $\mathbf{z}'$.
  Action: Samples $m \in \{0, 1\}$ using the absolute expectation $P_0 = 0.5$. Anchors the runtime parities $p_{X,k_p} \leftarrow p_{Z,k_p}$ and $p_{Z,k_p} \leftarrow m$. Appends m to sequence $\mathcal{M}$. The complex array a and global magnitude γ are invariant.

- `OP_MEAS_INTERFERE`**:** (Maps to Case 3b)
  Payload: Phase ω', pivot index k_p, aligned mask $\mathbf{z}'$.
  Action: Evaluates linear branch interference across the active elements of the unnormalized array a to calculate the structural weight p_0. Calculates absolute probability P_0 and samples outcome m. Folds the array a over itself (halving the active logical dimension) via division-free addition. Defers normalization to the global tracker via $\gamma \leftarrow \gamma/\sqrt{2P_m}$. Anchors parities $\mathbf{p}_X, \mathbf{p}_Z$ and appends m to $\mathcal{M}$.

- `OP_MEAS_DIAGONAL`**:** (Maps to Case 3c)
  Payload: Phase ω', pivot index k_p, aligned mask $\mathbf{z}'$.
  Action: Evaluates the unnormalized structural weight p_0 of branches matching the deterministic phase constraint. Calculates absolute probability P_0 and samples outcome m. Filters the array a by the valid boolean slice v without division. Defers normalization to the global phase scalar via $\gamma \leftarrow \frac{\gamma}{\sqrt{P_m}} \cdot (-1)^{p^{\text{tmp}}_{X,k_p} \cdot v}$. Anchors parities $\mathbf{p}_X, \mathbf{p}_Z$ and appends m to $\mathcal{M}$.

- `OP_MEAS_STATIC`**:** (Maps to Case 3d)
  Payload: Phase ω, static mask $\mathbf{z}$.
  Action: Evaluates outcome m against the current vacuum parities $\mathbf{p}_Z$. Appends m to $\mathcal{M}$. The array a and tracking scalar γ are invariant.

- `OP_NOISE`**:**
  Payload: Phase ω, masks $\mathbf{x}, \mathbf{z}$, error probability p_E.
  Action: Queries the continuous random variable $\xi^{(t)}$ to generate the indicator $b = \mathbf{1}_{\xi^{(t)} < p_E}$. Updates global phase γ via the indicator polynomial $\gamma \leftarrow \gamma \cdot \left[(1-b) + b \cdot \omega(-1)^{\mathbf{z}\cdot\mathbf{p}_Z}\right]$ and shifts parities $\mathbf{p}_Z \leftarrow \mathbf{p}_Z \oplus (b \cdot \mathbf{x})$ and $\mathbf{p}_X \leftarrow \mathbf{p}_X \oplus (b \cdot \mathbf{z})$.

- `OP_FEEDBACK`**:**
  Payload: Phase ω, masks $\mathbf{x}, \mathbf{z}$, classical condition index k.
  Action: Queries the memory sequence $m^{(k)} \in \mathcal{M}$ to set $b = m^{(k)}$. Updates global phase via the indicator polynomial $\gamma \leftarrow \gamma \cdot \left[(1-b) + b \cdot \omega(-1)^{\mathbf{z}\cdot\mathbf{p}_Z}\right]$ and shifts parities $\mathbf{p}_Z \leftarrow \mathbf{p}_Z \oplus (b \cdot \mathbf{x})$ and $\mathbf{p}_X \leftarrow \mathbf{p}_X \oplus (b \cdot \mathbf{z})$.

A Concrete 2-Qubit State Walkthroughs

To ground the abstracted mathematical update rules, this section walks through the complete evaluation of a small 2-qubit state. We construct a single unnormalized "Active Superposition" state, and then use it as a static branching point to independently evaluate each of the four measurement subcases. In the presentation below, bitstrings are indexed with the lowest index on the left, e.g. the bit for qubit 1 is the leftmost bit in the string.

To maximize readability, we assume runtime parities $\mathbf{p}_X = \mathbf{0}, \mathbf{p}_Z = \mathbf{0}$ initially. Because the global norm relies on the Deferred Normalization logic, we explicitly track the global parameter γ ensuring true absolute probabilities are evaluated accurately alongside the unnormalized structural components.

A.1 Initialization and Clifford Conjugation

We initialize the state to the vacuum $|00\rangle$. The geometric frame starts in the canonical computational basis, the active branch mask is empty, and the tracking variables initialize to 1:

$$\mathcal{S}^{(0)} = \langle +Z_1, +Z_2 \rangle, \quad \mathcal{D}^{(0)} = \langle +X_1, +X_2 \rangle$$
$$\mathbf{b} = 00, \quad a_{00} = 1, \quad \gamma^{(0)} = 1.0$$

We apply a Clifford Hadamard gate to the first qubit ($U = H_1$). By Clifford Conjugation rules, we apply $H_1 P H_1$ to every generator. The geometry updates, but the probabilistic tuple passes through unchanged:

$$\mathcal{S}^{(1)} = \langle +X_1, +Z_2 \rangle, \quad \mathcal{D}^{(1)} = \langle +Z_1, +X_2 \rangle$$
$$\mathbf{b} = 00, \quad a_{00} = 1, \quad \gamma^{(1)} = 1.0$$

A.2 Non-Clifford Dimensionality Expansion

We apply the non-Clifford rotation $T_1 \propto [I - i\tan\left(\frac{\pi}{8}\right) Z_1]$ to the state.

1. **Extraction:** We evaluate the physical operator Z_1 against the current frame. Because $d_1 = +Z_1$, the map trivially yields $\mathbf{x} = 10, \mathbf{z} = 00$ with $\omega' = +1$.

2. **Expansion:** Because $\mathbf{b} = 00$, the operator relies on dormant axes: $\mathbf{x}_{\text{dorm}} = 10$. We geometrically update the active rank to $\mathbf{b} \leftarrow 00 \vee 10 = 10$. The complex array expands, conceptually allocating padded bits: $a_{00}^{\text{pad}} = 1, a_{10}^{\text{pad}} = 0$. In practice, we've pre-allocated this space so no work is needed.

3. **Mixing:** The dynamic phase evaluates uniformly: $\Phi'(\boldsymbol{\alpha}) = (+1)(-1)^0 = +1$ for all $\boldsymbol{\alpha}$. Mixing a^{pad} against the geometric shift $\mathbf{x}' = 10$ yields the unnormalized branch amplitudes:

$$a_{00} = 1$$
$$a_{10} = -i\tan\left(\tfrac{\pi}{8}\right)$$

   The global scalar tracker updates as $\gamma \leftarrow 1.0 \cdot e^{i\pi/8}\cos\left(\frac{\pi}{8}\right) = e^{i\pi/8}\cos\left(\frac{\pi}{8}\right)$.

   (Note: The true absolute state norm is preserved: $|\gamma|^2 \sum |a|^2 = \cos^2(\frac{\pi}{8})[1 + \tan^2(\frac{\pi}{8})] = \cos^2(\frac{\pi}{8})\sec^2(\frac{\pi}{8}) = 1.0$*.)*

In the standard computational basis our state is now $\frac{1}{\sqrt{2}}\left(|00\rangle + e^{i\pi/4}|10\rangle\right)$.
(We now have an unnormalized active mask $\mathbf{b} = 10$*. We branch from this specific state to independently demonstrate the four distinct measurement scenarios.)*

A.3 Case 3a: Random Pivot / Vacuum Collapse

Action: Measure $P = Z_1 X_2$.

1. **Extraction:** In our frame $\mathcal{T}^{(1)}$, $d_1 = +Z_1$ and $d_2 = +X_2$, so $P = +d_1 d_2$. The extraction map yields $\omega' = +1$, $\mathbf{x} = 11$, $\mathbf{z} = 00$.

2. **Intersection:** With $\mathbf{b} = 10$, we have $\mathbf{x}_{\text{act}} = 10$ and $\mathbf{x}_{\text{dorm}} = 01$. The measurement acts outside the active superposition. We select the dormant bit $k_p = 2$ as the pivot.

3. **Rowsum:** We apply $\Lambda_{\text{rowsum}}(1, 2)$ to the geometry:

$$s_1 \leftarrow s_1 s_2 = +X_1 Z_2$$
$$d_2 \leftarrow d_2 d_1 = +Z_1 X_2$$

   The *Zero-Cost Dormant Property* ensures our complex array a is untouched by this compression.

4. **Pivot:** We now recompose P, which is now d_2 in the new geometry. We have support only over a_{00} and a_{10}, so there is no interference on this second qubit. The outcome is thus a 50-50 coin-flip. After we get the outcome m, we apply Λ_{pivot} to replace $s_2 \leftarrow P = Z_1 X_2$, and demote the old $s_2 = Z_2$ to the new d_2. d_1 remains Z_1.

A.4 Case 3b: Interfering Shrink

Action: Measure $P = Z_1$.

1. **Extraction:** Because $d_1 = +Z_1$, extraction yields $\omega' = +1$, $\mathbf{x} = 10$, $\mathbf{z} = 00$.

2. **Intersection:** With $\mathbf{b} = 10$, we partition $\mathbf{x}_{\text{act}} = 10$ and $\mathbf{x}_{\text{dorm}} = 00$. The measurement acts linearly within the active superposition. We select the active bit $k_p = 1$ as the pivot.

3. **Update:** The active dimension predictably halves: $\mathbf{b} \leftarrow 00$. The VM folds the state over the 1st geometric axis, calculating the relative unnormalized structural weight p_0:

$$p_0 = \tfrac{1}{2}\left|a_{00} + a_{10}\right|^2 = \tfrac{1}{2}\left|1 - i\tan\left(\tfrac{\pi}{8}\right)\right|^2 = \tfrac{1}{2}\sec^2\left(\tfrac{\pi}{8}\right) = \frac{1}{2\cos^2\left(\frac{\pi}{8}\right)}$$

   To evaluate the true absolute probability P_0 for sampling, the VM evaluates p_0 against the global magnitude ($|\gamma|^2 = \cos^2(\frac{\pi}{8})$):

$$P_0 = p_0 \cdot |\gamma|^2 = \left(\frac{1}{2\cos^2\left(\frac{\pi}{8}\right)}\right) \cdot \cos^2\left(\tfrac{\pi}{8}\right) = 0.5$$

   (Note: This matches the physical textbook expectation for measuring Z on $T\,|+\rangle$, avoiding divisions over the runtime array).

   Observing outcome m, the single remaining branch is derived via division-free addition:

$$a_{00} \leftarrow a_{00} + (-1)^m a_{10} = 1 - i(-1)^m \tan\left(\tfrac{\pi}{8}\right)$$

   To preserve the physical state norm post-measurement, Deferred Normalization updates the global magnitude instead of the array:

$$\gamma \leftarrow \frac{\gamma}{\sqrt{2P_m}} = \frac{e^{i\pi/8}\cos(\pi/8)}{\sqrt{1}} = e^{i\pi/8}\cos\left(\tfrac{\pi}{8}\right)$$

   The geometric pivot operation subsequently demotes $k_p = 1$ to anchor the new probabilistic vacuum.

A.5 Case 3c: Diagonal Shrink

Action: Measure $P = X_1$.

1. **Extraction:** Because $s_1 = +X_1$, extraction maps $\omega' = +1$, $\mathbf{x} = 00$, $\mathbf{z} = 10$.

2. **Intersection:** $\mathbf{x} = \mathbf{0}$, meaning there is no transversal mixing shift. But because $\mathbf{b} = 10$, we find $\mathbf{z}_{\text{act}} = 10$. The measurement acts diagonally across the active superposition.

3. **Update:** The dynamic phase evaluates as $\Phi'(\boldsymbol{\alpha}_{00}) = +1$ and $\Phi'(\boldsymbol{\alpha}_{10}) = -1$. The unnormalized structural expectation evaluates to $p_0 = |a_{00}|^2 = 1.0$.

4. **Filter:** Integrating the scalar, the true absolute probability evaluates to $P_0 = p_0 \cdot |\gamma|^2 = 1.0 \cdot \cos^2(\frac{\pi}{8}) \approx 85.3\%$ (matching the physical reality of measuring X on $T\,|+\rangle$).

   Assuming the VM samples outcome $m = 0$, the active dimension correctly halves ($\mathbf{b} \leftarrow 00$). The valid boolean slice $v = 0$ filters the array without division: we discard a_{10} and keep $a_{00} = 1$. Evaluating Deferred Normalization scales the physical state limits:

$$\gamma \leftarrow \frac{\gamma}{\sqrt{P_m}} = \frac{e^{i\pi/8}\cos(\pi/8)}{\cos(\pi/8)} = e^{i\pi/8}$$

A.6 Case 3d: Deterministic Filter

Action: Measure $P = Z_2$.

1. **Extraction:** Because $s_2 = +Z_2$, extraction explicitly maps $\omega' = +1$, $\mathbf{x} = 00$, $\mathbf{z} = 01$.

2. **Intersection:** With $\mathbf{b} = 10$, both $\mathbf{x}$ and $\mathbf{z}_{\text{act}}$ evaluate to $\mathbf{0}$. This part of the state is the current vacuum state, which is an eigenstate of $P^{(t)}$.

3. **Update:** The VM bypasses the array and probabilistic random number generator. It evaluates the vacuum parity eigenvalue $\omega'(-1)^{\mathbf{z}\cdot\mathbf{p}_Z} = +1(-1)^0 = +1$, yielding $m = 0$. Neither the array, the active mask, nor the global magnitude are altered.

A.7 Conditional Pauli Noise

Action: Resume from the active multi-branch superposition established in A.2 ($\mathbf{b} = 10$, $a_{00} = 1$, $a_{10} = -i\tan\left(\frac{\pi}{8}\right)$, $\gamma = e^{i\pi/8}\cos\left(\frac{\pi}{8}\right)$, $\mathbf{p}_X = 00$, $\mathbf{p}_Z = 00$). Let the environment apply a stochastic Y_1 Pauli error with continuous probability p_E.

1. **Extraction:** In the geometric frame $\mathcal{T}^{(1)}$, $s_1 = +X_1$ and $d_1 = +Z_1$. The physical Pauli factors as $Y_1 = iX_1Z_1 = is_1d_1$. By canonical anti-commutation ($s_1d_1 = -d_1s_1$), this maps to $-id_1s_1$. The extraction map yields $\omega = -i$, $\mathbf{x} = 10$, $\mathbf{z} = 10$.

2. **Evaluation:** We query the random variable ξ to determine the outcome b. If b is 1, we apply the gate.

3. **Update:** The global phase absorbs the extracted scalar mapping:

$$\gamma \leftarrow \gamma \cdot \left[(1-b) + b \cdot \omega(-1)^{\mathbf{z}\cdot\mathbf{p}_Z}\right] = e^{i\pi/8}\cos\left(\tfrac{\pi}{8}\right) \cdot (-i)(-1)^0 = -ie^{i\pi/8}\cos\left(\tfrac{\pi}{8}\right)$$

   The parities shift via bitwise evaluations:

$$\mathbf{p}_Z \leftarrow 00 \oplus 10 = 10$$
$$\mathbf{p}_X \leftarrow 00 \oplus 10 = 10$$

4. **Physical Consequence:** Substituting these updated parities into the physical state equation defines the new local physical generators as $\bar{s}_1 = (-1)^1 X_1 = -X_1$ and $\bar{d}_1 = (-1)^1 Z_1 = -Z_1$.

   This shifts the physical base vacuum state from $|+\rangle_1 |0\rangle_2$ to $|-\rangle_1 |0\rangle_2$. Evaluating the invariant complex array a over this shifted physical vacuum and negated destabilizer yields:

$$
\begin{aligned}
|\psi\rangle &= \gamma \Big[a_{00} (\bar{d}_1)^0 |-\rangle_1 |0\rangle_2 + a_{10} (\bar{d}_1)^1 |-\rangle_1 |0\rangle_2 \Big] \\
&= -ie^{i\pi/8} \cos\!\left(\tfrac{\pi}{8}\right) \Big[1 |-\rangle_1 |0\rangle_2 - i \tan\!\left(\tfrac{\pi}{8}\right) (-Z_1) |-\rangle_1 |0\rangle_2 \Big] \\
&= -ie^{i\pi/8} \cos\!\left(\tfrac{\pi}{8}\right) \Big[|-\rangle_1 |0\rangle_2 + i \tan\!\left(\tfrac{\pi}{8}\right) |+\rangle_1 |0\rangle_2 \Big] \\
&= -ie^{i\pi/8} \cos\!\left(\tfrac{\pi}{8}\right) |-\rangle_1 |0\rangle_2 + e^{i\pi/8} \sin\!\left(\tfrac{\pi}{8}\right) |+\rangle_1 |0\rangle_2
\end{aligned}
$$

   This equates exactly to applying a dense physical Y_1 matrix against the original multi-branch superposition state $(Y_1 T_1 |+\rangle_1 |0\rangle_2)$, achieved here at zero computational array cost.